2 9 MAR 2019

Roll No. _____

B.C.A Sem-3 Examination

Subject Code: 20631

Data and File Structure

Total: 70 Marks **Time: 2 ½ Hrs**

Q – 1 Answer the following questions.

1. Define data structure in detail 07
2. Explain Row Major and Column Major Array Representation. 07

OR

Q – 1 Answer the following questions.

1. Explain Insertion sort with example. 07
2. Write an algorithm for Merge Sort. 07

Q – 2 Answer the following questions.

1. Write note on Doubly link list 07
2. Write an algorithm for the following operations of doubly link list 1) insert 2) update first 3) delete last 07

OR

Q – 2 Answer the following questions.

1. Write a code for doubly link list's following functions create() and display() 07
2. Give various Advantage and Disadvantage of Doubly link list 07

Q – 3 Answer the following questions.

1. Write a note on stack. Also state the applications of stack. 07
2. Write an algorithm for push, pop and display in stack. 07

OR

Q – 3 Answer the following questions.

1. Convert following to postfix 07
   A+(B*C-(D/E-F)*G)*H
2. Write a Stack Program using array. 07

Q – 4 Answer the following questions.

1. Wrie note on queue and its types 07
2. Write algorithm of create and display element in simple queue. 07

OR

Q – 4 Answer the following questions.

1. Write a program of simple queue(insert, delete, update, display). 07
2. Explain Circular Queue using array with example. 07

Q – 5 Answer the following questions.

1. Explain Binary Tree and its types. 07
2. Explain following terms 07
   Node, Parent node, Child node, Sibling node.

OR

Q – 5 Answer the following questions.

1. Explain BST traversal techniques in detail. 07
2. Write note on graph. 07